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Professor Darius Ceglarek and Members of the Selection Committee
Advanced Manufacturing Processes and Systems (AMPS) Research Group
WMG, University of Warwick
Coventry CV4 7AL, United Kingdom

Dear Professor Ceglarek and Members of the Selection Committee,

Application for Assistant Professor in Artificial Intelligence for Manufacturing Systems, 77978-0526

I am writing to apply for the **Assistant Professor in Artificial Intelligence for Manufacturing Systems** post (**77978-0526**) in the **Advanced Manufacturing Processes and Systems (AMPS) Research Group** at WMG, University of Warwick. I am currently a Research Assistant at the **Power Electronics and Machines Centre (PEMC), University of Nottingham**, having completed my **PhD in Electrical Engineering at Southeast University (SEU), Nanjing**, in September 2025 with the dissertation *Multi-physics Analysis and Design of Hairpin Winding PM Traction Machines in Electric Vehicles*. My research has developed AI-driven multi-physics design, digital-twin modelling and advanced manufacturing methods for high power-density electrical machines, and I would be glad to contribute to AMPS's research and teaching at the intersection of artificial intelligence, digital manufacturing, and intelligent industrial systems.

A central thread of my work is **AI/ML for manufacturing**, anchored by my ICEM 2026 paper *A Deep Learning Based Magnetic Field Surrogate Modeling Method for Interior Permanent Magnet Machine in EVs*. The paper introduces an **implicit physics-knowledge embedding** approach in which a deep neural network learns to map machine design parameters and rotor position to the full **magnetic vector potential distribution**, with a Fourier feature layer for the spatial coordinates. Conventional post-processing then recovers torque, flux linkage and iron loss. The model reaches $\mathbf{R^2 = 0.9996}$ for average torque and **0.9730** for iron loss on an unseen validation set, while reducing inference time by **more than 94%** against direct finite-element analysis. The surrogate sits on an **in-house Python finite-element code package** I have developed, covering electromagnetic, structural-modal and mechanical-stress analyses. Together they form a **physics-grounded digital twin** of the machine that supports rapid what-if studies, design-space exploration and AI-assisted optimisation, and is currently being applied to magnet-free synchronous machine work at PEMC. This combination of physics-based simulation, deep-learning surrogates and open, scriptable software is directly relevant to AMPS's interests in digital twins, manufacturing analytics and software-enabled industrial systems.

My research record to date includes **22 journal papers, of which 17 are in IEEE Transactions** (across *IEEE Trans. Industrial Electronics*, *Transportation Electrification*, *Industry Applications*, and *Energy Conversion*), 18 peer-reviewed conference papers at IEEE IEMDC, IEEE ECCE, ICEM and ICEMS, and 9 patents. The manufacturing-engineering content includes **novel n-layer flat-wire and hairpin winding design** with conductor-layout optimisation and tolerance-aware mechanical design, **high-fidelity lumped-parameter and fluid-solid coupled oil-cooling models** validated against prototype measurements, and **multi-objective multi-physics optimisation** coupling electromagnetic, thermal and mechanical objectives. This work has been recognised by the **2025 IEEE TEC Electric Vehicle Design Competition (1st Prize)**, the **2025 IEEE Trans. Transportation Electrification Prize Paper Award (2nd Prize)**, the **1st Prize in the IEEE IAS CMD Thesis Contest (non-PhD category)**, the **CIEEC Best Paper Award (2023)**, and the **Jiangsu Province Outstanding Master Degree Thesis Award (2022)**. I also serve as **Co-Organiser and Co-Chair of an ICEM 2026 Special Session** on *Next-Generation Electrified Transport* in Funchal, Madeira, Portugal.

I have also gained substantive **industrial development experience** through a multi-year collaboration with **Inovance**, a world-leading EV traction-machine supplier, on two oil-cooled PM traction machines: an **18 000 rpm 113 kW A0-class machine** (138 Nm, 7.78 kW/kg, 95.3% peak efficiency) and a **20 000 rpm 270 kW A-class machine** (380 Nm, 7.87 kW/kg, 97.4% peak efficiency). Each was taken through a full industrial cycle of multi-physics design, engineering drawings and tolerance specification for high-reliability manufacture, and

AVL dynamometer characterisation. This combined academic and industrial background should support future funding applications across UK research councils and industrial partners.

I would be glad to contribute to teaching across WMG's portfolio at undergraduate, MSc, Executive Education and Masterclass level, in areas such as **AI for manufacturing engineers, finite-element principles and FEA-based design, digital-twin laboratories, multi-physics machine design, and embedded systems for project-based learning**. During my doctoral studies I served as teaching assistant on the SEU course *Integrated Practice of Electrical Machine Systems*, preparing tutorial material, leading Q&A, reviewing project proposals, and contributing mark recommendations. My toolchain spans MATLAB/Simulink, ANSYS Maxwell and Mechanical, Python and PyTorch, and C/C++ for TI C2000 and STM32, complemented by hands-on power-electronics prototyping. I would also be glad to undertake **PCAPP** and **Higher Education Academy Fellowship** in due course.

For background, my doctoral supervisor, **Prof. Zhongze Wu**, held a Research Fellow position in electrical machines at WMG, University of Warwick from January 2017 to August 2018, which has given me a long-standing familiarity with WMG's research culture through his mentorship.

My CV and detailed responses to the Person Specification are attached. I would be glad to discuss my research and teaching plans further at interview.

Yours sincerely,

Wentao Zhang